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Program: Letter grade converter

```
#include <iostream>
using namespace std;

int main(){
    int marks;
    cout<< "Letter grade converter\n";
    cout<<"Enter numerical grade (1-100): ";
    cin>>marks;

    if (marks <=100 && marks >= 88){
        cout<<"Letter grade: A\n";
    }
    if (marks <=87 && marks >= 80){
        cout<<"Letter grade: B\n";
    }
    if (marks <=79 && marks >= 67){
        cout<<"Letter grade: C\n";
    }
    if (marks <=66 && marks >= 60){
        cout<<"Letter grade: D\n";
    }
    if (marks <=59 && marks >= 0){
        cout<<"Letter grade: F\n";
    }
    if (marks <0){
        cout<<"Numerical grade cannot be negative.\n";
    }
    else{
        cout<<"Numerical grade is too large.\n";
    }
    cout<<"Bye!";

    return 0;
}
```

Program: Shape calculator

```
#include <iostream>
```

```
#include <cmath>
```

```
using namespace std;
```

```
int main() {
```

```
    char shape;
```

```
    double area;
```

```
    cout << "Enter the shape you want to calculate the area of (C = Circle, R = Rectangle, S = Square): ";
```

```
    cin >> shape;
```

```
    switch (shape) {
```

```
        case 'C':
```

```
            double radius;
```

```
            cout << "Enter the radius of the circle: ";
```

```
            cin >> radius;
```

```
            area = M_PI * pow(radius, 2);
```

```
            break;
```

```
        case 'R':
```

```
            double width, height;
```

```
            cout << "Enter the width of the rectangle: ";
```

```
            cin >> width;
```

```
            cout << "Enter the height of the rectangle: ";
```

```
            cin >> height;
```

```
            area = width * height;
```

```
            break;
```

```
        case 'S':
```

```
            double side;
```

```
            cout << "Enter the side length of the square: ";
```

```
            cin >> side;
```

```
            area = pow(side, 2);
```

```
            break;
```

```
        default:
```

```
            cout << "Invalid shape entered. Please enter C, R, or S.";
```

```
            return 0;
```

```
    }
```

```
    cout << "The area of the shape is: " << area << endl;
```

```
    return 0;
```

```
}
```

Program: Pig Dice rules

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {
    // File name
    string fileName = "rules.txt";

    // Open file
    ifstream inFile;
    inFile.open(fileName);

    // Check if file is open
    if (!inFile) {
        cout << "Unable to open file: " << fileName << endl;
        return 1;
    }

    // Read file line by line
    string line;
    while (getline(inFile, line)) {
        cout << line << endl;
    }

    // Close file
    inFile.close();

    return 0;
}
```